

Communication Report

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Introduction

This report covers communications and related work done during summer as well as wintering period from 5 Dec 92 to 26 Jan 94. The report aims to highlight the major events as well as offer suggestions to further improve the communication in future.

Summer Period

Communication Trials Prior to Departure

Communication trials with COMCEN(D) and Maitri were conducted satisfactorily on 2 and 3 Dec 92 using ship-borne communication equipments. The frequency used were as follows:

- a) COMCEN/ *THULELAND* - 14450/16941 - 1000Z to 1130Z
- b) COMCEN/ MAITRI - 14450/16941 - 1130Z to 1300Z
- c) MAITRI/ *THULELAND* - 16941/12528 - 1300Z to 1400Z

Communication During Passage

(a) The ship maintained HF communication with COMCEN(D) and Maitri satisfactorily throughout the passage. Communication with Maitri was also established and maintained on 16941 kHz using ship borne Maitri in FEC mode from 8 Dec 92. The ship experienced difficulty in communicating with Delhi after crossing 67°S. Thereafter, the signal traffic for Delhi was routed through Maitri.

(b) The readability of Broadcast BN was good upto 8°S. Thereafter VTG 8/9 were heard satisfactorily with occasions of interference.

Communication at Antarctica

HF R/T contact was established with Maitri on 26 Dec 92 satisfactorily. The frequencies used were 4460 and 6203 kHz. Hereafter, a constant communication was maintained with Maitri on scheduled or as on required basis. Spot frequency 4460 kHz was found to be better at a distance of about 130 km from Maitri between 0600 and 2000 hrs GMT daily. The ship arrived in Antarctica on 28 Dec 92.

Communication between Ship and Halos

VHF communication between ship-helicopters and Maitri- helicopters was very good throughout the summer period. The frequency in use was 127 MHz.

Communication with Dakshin Gangotri

Dakshin Gangotri (DG) station was activated from time to time and on as required basis. Communication between ship-DG and Maitri-DG had been very good.

Performance of Communication Equipment

The performance of ship borne communication equipments had been very good throughout. However, 1 kW main HF transmitter developed defect after arrival in Antarctica. On defection by the POELR it was established that Thyristor in servo-motor of tuning circuit was defective. The ship carried no spares to enable to carry out necessary repairs.

Disruption of Communication

(a) The communication with Maitri and Delhi was disrupted on 28 Dec. 92, 2 and 3 Jan. 93 due to severe magnetic storm in the area.

(b) Although HF/T could meet the expedition requirements yet the signals used to fade away occasionally due to near nonexistence of HF ground wave on ice shelf. The link operated at 400 W power output from ship and 125 W from Maitri and DG.

Commissioning of New INMARSAT at Maitri

A new INMARSAT equipment along with its accessories was embarked at Goa for installation at Maitri. The same was airlifted to Maitri on 2 Jan., the very second day of flying. The dome antenna being in an oversize container could be airlifted only on 5 Jan. due to very bad weather. After the installation

and fitment on antenna the INMARSAT was successfully tested by Maitri communication team on 17 Jan. 93.

On Job Training

Communication and electrical personnel were sent to Maitri in rotation for familiarisation in a phased manner before formal handing/taking over of station. This helped a great deal in getting a feel of station and learning the working pattern at Maitri and DG.

Communication Equipment brought from India

The following new/repared equipments were supplied by the DOD and brought to Maitri:

(a) INMARSAT with accessories	1	New
(b) PR 10-8 Walkie Talkie sets	4	Repaired
(c) Dual unit charger for PR 10-8	4	"
(d) Tech.H/B Exciter T1770/MMT-4000	1	"
(e) TX/MP sets (VHF) PWSL	3	"
(f) MP 10-25 (VHF) PWSL	1	"
(g) Rear Panel 100 W Amplifier	2	"

Restriction on Use of Ship's Transmitters

To enable ship receive weather charts on facsimile without local disturbances the use of ship's transmitter was forbidden at following hours (UTC) for an approximately duration of 10 to 15 minutes each time.

0445, 0520, 0800, 1000, 1445, 1700 & 2200 hrs.

Winter Period

HF Communication

HF Link with Delhi

(a) The bulk of messages between Maitri and the Department of Ocean Development (DOD) was handled by using HF link between Maitri and the Naval Headquarters. Technical and other details of this HF link are placed at appendix 'A'.

(b) The HF link worked very well and was able to meet the requirement of the expedition. While the overall performance of the link was excellent, the day to day performance depended much on HF ionospheric propagation conditions

which keep changing due to magnetic storms, proximity to the pole and various other factors affecting the HF propagation. The propagation condition were poor during Polar night (May to July) and during the days of severe geomagnetic activities (Magnetic storms). The HF communication was also affected during strong blizzards/heavy snowfalls due to build up of heavy static charges in the station. The performance of the equipment is adversely affected when operated during such periods. The details are as follows :

- | | | | |
|------|-----------------------------------|---|----|
| (i) | Total number of no contact days | : | 32 |
| (ii) | Total number of poor contact days | : | 19 |

Most of the no contact days and poor contact days were during Polar night and the rest were spread over the entire period.

(c) As most of the messages were of routine and private nature, these used to be transmitted over HF same day or the very next day if these remained pending for any of above reasons. Passing of such messages over the satellite link was never resorted to as the same was not considered necessary.

Message Traffic Pattern

Most of the message were of private nature from members to their families/friends etc. and vice versa. These messages were passed and received on HF. The official traffic from Maitri to DOD and IIG Bombay, IMD Delhi, RDE Pune, Naval Headquarters, Army Headquarters etc. was passed on HF via DOD. Certain official messages of importance and urgent nature were passed over Satellite link also duly authorised by the leader of the expedition at his discretion. It was observed that the number of outgoing messages from Maitri was five times higher than what was received at Maitri.

Personal Messages

Personal messages formed bulk of the traffic handled (both IN and OUT). It was observed that wintering members here wrote more messages than what they received from their families and friends. This sometimes led to an impression that their mail were not despatched by DOD to their families properly. But this proved to be wrong. In fact, it is learnt that DOD despatches the mail correctly and promptly but in most of the cases the families/friends of members do not respond promptly to the messages.

Cost Savings

A total of approx. 1325 messages were handled by this team using HF link. Out of this 1075 messages were from Maitri to DOD/NHQ and 250 messages from DOD/NHQ to Maitri. The total number or words works out to be approx.

2,83,000. If the same length of words were to be passed/received over Satellite link the cost would have worked out to be approx. Rs. 3,50,000. In addition to above, about 150 messages were exchanged between Maitri and MV *Stepan Krishninnikov* from the time the ship left Goa.

Satellite Communications

Maitri is linked to the rest of the world via Satellite Communication terminal of INMARSAT (International Maritime Satellite) system. This was mainly utilised for sending and receiving official telex messages as well as for making official/private telephone calls. In addition to Telex and telephone the station has facilities of receiving and sending Facsimile (Fax) messages both official and private. During the team's wintering this system has been very popular and was sufficiently used. Members sent their handwritten/typed messages to their near and dear ones. Even neat sketches, drawings, greetings etc. were also sent duly drawn on plain paper with the help of computer. Time utilized for sending Fax messages is counted towards individual's telephone time.

One Satcom terminal in Radio room is lying non-operational since May 92 for want of spares. The spares for the same have not been received at Maitri till todate. Another Satcom terminal fitted in NPL Hut was working till mid April 93. It became non-operational due to erratic power supply from the source and it was found that teleprinter, stabilizer and cards of the main unit were damaged/burnt. The spares have been demanded by this team and system will start working once again.

Satcom terminal presently working in Radio room was installed in Jan 93 and is working fine till today in all three modes, *i.e.*, Phone/FAX/Telex, except one day when due to heavy static charge built up in station and a card was rendered non-operational in its power supply unit. Not a single spare for this Satcom terminal was supplied by the suppliers (The entire terminal, accessories were directly received at Goa from abroad). The terminal was repaired very same day by replacing a burnt diode by cannibalisation and system restored. Sufficient spares for this system have also been demanded which would be coming through XIII expedition.

Weather data from Maitri was collected by the India Meteorological Department (IMD) at 0000, 0600, 1200 and 1800 UTC hrs daily on telex.

In view of very high cost of Satellite communication, the usages of this system has been very judicious: Most of the messages from Maitri to DOD/NHQ, both official and private were sent on HF teletype/CW. However,

it has been observed that most of the messages from DOD to Maitri came on Telex.

Convoy/Camp Communications

Communication with 13 convoys conducted by this team from Feb 93 to Nov 93 had been very good. Three convoy vehicles have been fitted with VHF antenna by this team for use with VHF dittel Tx/Rx. In fact only two vehicles could be provided with VHF sets since station has only three such sets. One always work at main station in Radio room. Three such sets were backloaded by our predecessors for repairs. An effective communication has been achieved by this team from convoy to Maitri and vice versa upto a distance of 75 to 80 km. HF communication with DG also had been good. HF communication between Maitri and convoy vehicles has not been found effective due to absence of HF ground wave on ice surface hence discarded.

Necessity was also felt by the convoy commander and expedition leader to have additional long range walkie talkies (8 to 10 km) for communication between the vehicles specially when operating at a distance and in bad weather conditions. The same has been recommended to DOD with a request to procure five such sets. Walkie talkies PR-10-8 practically have less range (1 to 1.5 km) when operated in cold weather and are not considered ideal for convoy communications.

VHF communication with scientists team camping in Schirmacher range has also been good. The VHF sets had been either used in Sankalp or in the vehicle with them.

Communications with Neighbouring Antarctic Stations

Daily communication was maintained with Russian station Novolazarevskya on 127 MHz and the German station, George Foster on 4460 kHz. A continuous listening watch on these two frequencies is maintained at Maitri and this station also maintained round the clock watch on their respective frequencies. During the winter period of 1993, the German station, George Forster was closed down formally. Communication with other German station Neumayer (about 700 km away) was also made whenever required and the propagation condition were found to be good. However, most of the stations have expressed their desire to work on Telex/Fax mainly perhaps due to English language problem.

Problems in Communication

Static Noise

Maitri has numerous metallic objects and riggings etc. in its vicinity. The station building also has metallic cladding in some places. All this metallic objects tend to acquire static electric charges and cause radio noise. This problem is made worse due to wooden building structure which is non-conductive and provides absence of electric ground for the station. This results in malfunctioning of sensitive radio equipments. It has also been observed that in the presence of static charges other scientific equipment also get affected and malfunction. At many occasion the equipment were damaged also. Therefore, to ensure noise free communication and better upkeep of costly electronic and sensitive equipments a proper radio ground is a necessity. It is recommended that a separate grounding should be considered for radio equipments. The effect of static charge has been felt mostly during Polar night/blizzards/heavy snow-fall.

Radio Aerials

At Maitri, we do not have proper antenna systems except for a Rhombic antenna for high power transmission. All other transmitting and receiving aerials are make shift and do not work at their peak efficiency. Basically it is the height of the aerial that matters here as Maitri is in a valley surrounded by hills and glacier. Even the present aerial can perform better if the mast on which they are installed are raised. For example, the very high frequency VHF antenna used for line of sight of communication with helicopter/ship does not pick up ship's transmission in the India Bay. But the moment helicopter gets airborne Maitri get in touch with it in one minute time. The India Bay is about 110 to 120 km from Maitri. A further rise in height of (VHF) antenna can prove a boon for this station as communication with convoy vehicles, camping parties at greater distance from Maitri can be maintained upto about 120 km very efficiently. Presently, the camping parties in Wohlthat mountain ranges are provided with HF equipment. They are unable to communicate with helos as helos do not carry HF equipments. The aim should be to create a VHF communication network during summer season where ships/helos/camp/Maitri can communicate with each other directly. Moreover, it is easier to install and carry VHF lighter equipments than bulky and heavy HF sets/power packs/ATU's and the aerials. Provision of better antenna with height advantage is a point which requires planning and proper consideration as no amount of better and powerful radio equipments will be of help when antenna system is

poor. We have observed that neighbouring Russian and German stations have very extensive antenna system on high mast.

Power Supply

It has been observed over a period of one year usage that main power supply in Maitri is erratic. The level of voltage fluctuations and transients is above the limit acceptable for radio equipments to function properly. In many cases this has resulted in equipment breakdowns and consequently avoidable manpower wastage in defect rectification. In fact, it happens because the supply provided to radio equipments is from the same source as that of other station equipments and electric appliances. Running of heaters, blowers, motors, electric kettles, iron etc. and their continuous on/off causes lot of fluctuations in station electric supply systems. It is recommended that a dedicated power transmitter can be considered by using a 30 KVA generator set. This will improve the efficiency and life of costly radio equipments.

Mutual Interference

The equipment density in Radio room is high. This small room has a one KW transmitter, two HF receiver, two VHF transreceiver, Satcom terminal with telephone, teletype, Fax and VDU etc. Two main 5 kW transmitter (HF) are also located within five metres of the Radio room. All this causes mutual interference. More space is required for Radio equipments. A separate hut immediately behind the station would be ideal for transmitters.

Equipments and Spares

Equipment Availability

Availability of equipments has been very good. Most defects were attended to immediately. However, the availability of spares for HF transmitter and HF receivers has not been quite adequate. Main HF transmitter (5 kW), HF receiver RS 512, one kW linear power amplifier etc. did not have sufficient spares to see through the whole wintering. Even the new Satcom system did not have a single spare with its initial supply. The radio auto teletype mode on main transmitter (5 kW) could not be supported for want of spares from the month of April/May itself. CW mode was used throughout thereafter for passing messages.

Repair Expertise

The communication team had adequate expertise in repair and maintenance of communication equipment and associated antennas used with them. Even the repairs to Satellite communication system were carried out successfully. Repairs to non-communication equipment like photocopying machine, were also successfully carried out. The team designed and rigged antennas on vehicles and in station and improved upon the HF and VHF communication. Helping hand was always given in maintaining other electronic gadgets in station whenever it was required. Some of the HF receivers earmarked for back loading last year and not back loaded due to shortage of time, were also repaired by this team without proper spares. One main 5 kW HF transmitter which was taken over working at half its power output, was repaired by this team and the transmitter worked throughout the year at its peak efficiency. Also, the new genset Aditya building was provided with intercom system through cannibalisation as no equipment was available in the station.

Improvements and Modifications

The RATT (Radio Auto Teletype) was exploited fully for sending/receiving messages till onset of winter. The fixed service working was resorted to CW mode due to two reasons. Firstly, the station did not have adequate spares for main 5 kW HF transmitter to support the RATT operation, and secondly, a minimum signal strength 3 is mandatory to have satisfactory printability of 2/3, which is not possible during the period May to August due to unfavourable HF propagation conditions. This team worked out in details the requirement of such spares and demanded the same.

Three HF receivers DEBEG actually planned for back loading by the IX wintering team have been locally repaired by this team and all are fully operational now. No spares for these equipments were available.

Adequate spares for new and old satcom terminals have been demanded. The new satcom system was supplied without any spares.

Two new receiving antennas have been fabricated and installed by this team as per required orientation and are giving very good performance.

Altogether 3 new antennas have been fabricated suitably to work on convoy vehicles and installed properly. They are working satisfactorily.

Main transmitter No. 2 of 5 kW capacity was taken over by this team performing at only 2.7 kW output. The same has been repaired and being used with full power output.

During Feb 92 to Feb 93 this team faced many blizzards and bad weather days. The strongest blizzard ever recorded at Maitri clicked 92 knots an hour. Everytime the high winds damaged some antenna or the other. 5 kW main Rhombic antenna/structure was badly damaged during a blizzard. All these defects were undertaken at the earliest opportunity and systems were restored.

This team proudly put on record the highest number of HF messages ever handled at Maitri so far. This had been possible only through ensuring availability of equipment at all times and sincere hard work put in by the members.

Recommendations

1. Two DRDO masts which have come up in the beam path of the main Rhombic antenna have degraded its performance. In this regard a case was taken up by the IX winter team with DOD through NHQ. Instructions in this regard have not been received so far. However, during X wintering the opinion of this team and DRDO representative was sought by the DOD in this regard. The joint opinion was that the DRDO masts should be relocated away from the present site, and it was conveyed to DOD by the leader of X wintering team. It was recommended that this case may be approved at the earliest and task be allocated as considered convenient.
2. An antenna mast is lying about 300 metres behind the station with proper basement made since 1988. This mast has not been erected so far since it was not included as a task for any summer party by the DOD. It is recommended that this task may be approved/included so that the mast can be used for rigging receiving antennas on this. Even this mast can be used to install a VHF antenna since this location will have height advantage over other masts around the station.
3. VHF antenna presently rigged on main building is working since commissioning of the station. It has been discussed with army engineers that the height of this pole can be increased by another 2 to 3 metres without any risk of its falling. If this could be done an increased range of about 10 to 20 km can be achieved in VHF communication. It is recommended that this task also may be considered and approved so that VHF communication with convoy vehicles and helicopters/ship can be further improved.
4. As brought out in the preceding paragraphs regarding separate RF earthing for Radio equipment, it is strongly recommended that this task may be considered on priority in order to improve communication and life of costly electronic/radio equipment.

Conclusions

Naval communication team has been able to meet the diverse communication requirements of station at all times and under all circumstances. Be it bad weather, snowfall or the raging blizzard for that matter. Work continued with full dedication in the best interest of the expedition and keeping in mind the best traditions of this fine service. The present set up and equipment at Maitri are considered adequate to meet the requirement of the expedition but there is always room for betterment and modifications in the overall interest of station and in keeping pace with the modern technology. It may be appreciated that it is the availability of trained personnel with equipment maintained to work at their peak efficiency, to achieve reliability and speed in communication. In a place like Maitri, it is thus, mandatory to keep adequate stock of equipments and their spares to maintain the required efficiency throughout the year. It is, therefore, recommended that the proposals forwarded by the team here with 16 months of experience under all sorts of adversities, may be considered to improve upon existing systems and make it a useful tool for our Scientists to achieve their objectives on this continent.

Acknowledgements

The communication team heartily thanks the members of X wintering who have contributed their bit directly or indirectly towards performing our duties satisfactorily in meeting the requirements of the expedition. Special thanks to the Army team who so readily provided us the technical/material support whenever and wherever it was required without which the accomplishment of tasks would not have been probably possible. Also, a morale support which anybody on this continent would look for to achieve the goal through one's task, came as blessings from Dr V.K. Dhargalkar, the leader, to whom this team would ever remain indebted. Under his able leadership and with personal touch everything that one may call "difficult" looked easy throughout the period of expedition.

Appendix 'A'

Details of HF link between Maitri and Naval Headquarters.

1. Frequencies
(A) 16941 kHz by Maitri for passing messages
(B) 14450 kHz by Delhi for passing messages
2. Mode of operation : RATT/CW
3. Working schedule : 1100 UTC to 1500 UTC (Approx) daily
except Sundays/Holidays
4. Equipment used : Punwire 5 kW RF 727 transmitter and HF
receivers RS 512/HS 4120 and other
associated equipment for RATT operation
5. Antennas used : Rhombic antenna for Transmitter and
dipole/inverted 'V' antenna for receivers